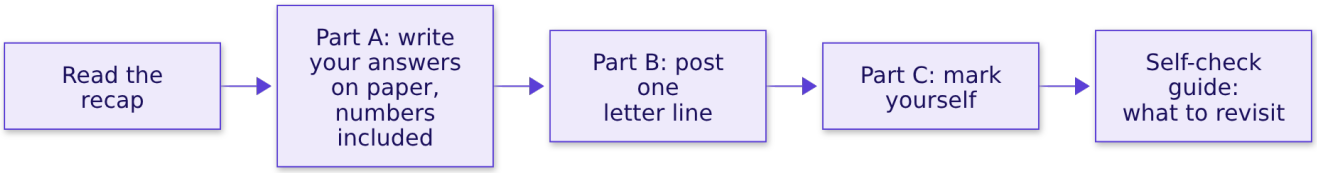
Practice set: regression and classification, framed first

Week 5, Monday. About two and a half hours, done alone, with the key at the end.

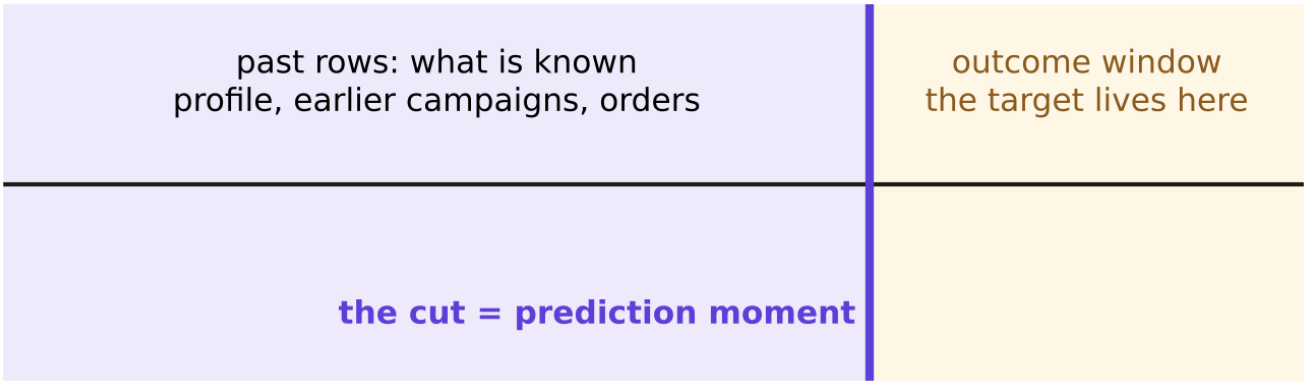
How do you use this set?

Part	What	Minutes
Recap	one page of pictures and rules	10
A	ten problems: three warm-ups, four medium, three interview grade	100
B	eighteen self-check questions	25
C	the key and the model solutions, read after you finish	15

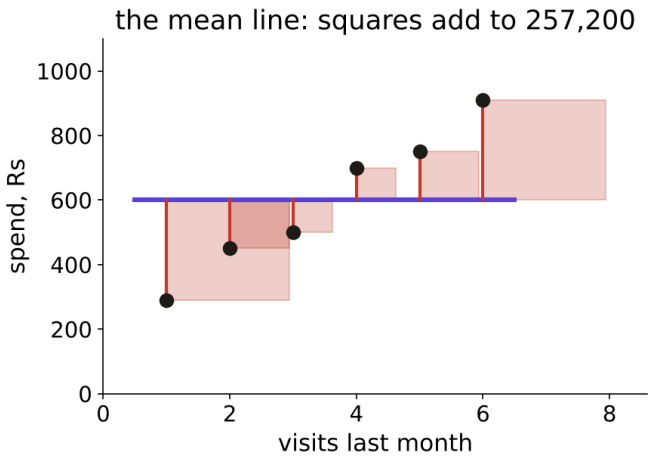


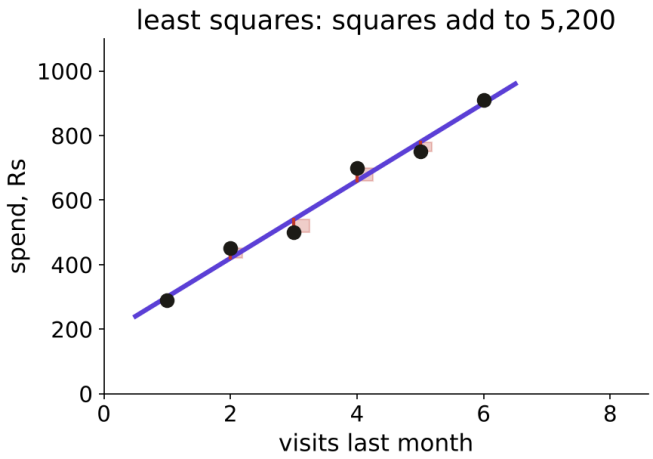
Each problem names the seat it is answered from: business owner, product manager, solution architect, senior ML engineer, implementation engineer, or QA and tester. Two problems are under-specified on purpose; a strong answer opens with the one question worth asking before any number.

What does the day come down to?



Idea	The rule	The condition that flips it
Query, rule or prediction	a model earns its place when the input is known now, the target later, and the decision moves with the score	a rule stays right until its misses cost more than a model would
Prediction time	every input comes from the left of the cut; the target from the window right of it	none: a column from the right of the cut is a leak
Three slices	train learns, validation chooses, test is scored once	a peeked test is validation now; cut a fresh slice
Baseline	majority class for "will they", the mean for "how much"	a cheap rule, where one exists, is the stronger bar
Linear regression	the least-squares line leaves the smallest sum of squared residuals; read the slope in the target's units	the slope is an association unless an experiment backs it
Logistic regression	`p = 1 / (1 + exp(-(a + b x)))`; sign is direction, `exp(b)` is the odds multiplier	one S holds one direction; a chance that rises at both ends breaks it
Residuals	a pattern is a shape the model missed	a flat plot vouches only for the data's range





Part A: ten problems

A1. Which of Rohan's questions needs a model? (warm-up, 5 minutes)

Seat: product manager. Rohan Desai, head of risk at Kalpa Financial Services, sends five questions (invented).

#	Rohan's question
1	How many loan applications did we decline in September?
2	Which approved borrowers will miss a payment in the next six months?
3	Decline any application with no PAN on file
4	How much will each borrower repay next quarter?
5	What is the average loan size in Hyderabad this year?

1. Sort each into query, rule or prediction.
2. For each prediction, say whether it is regression or classification.
3. For question 3, name one fact that would turn it into a prediction problem.

In the interview: "When is a business problem a machine learning problem? Use one of these five to show it."

A2. Does 99.4 percent beat doing nothing? (warm-up, 5 minutes)

Seat: senior ML engineer. A payments app's fraud team (illustrative, numbers invented) reports on its validation slice:

Validation slice	Rows
genuine payments	9,950
fraudulent payments	50
the model's accuracy	99.4 percent

1. What accuracy does "call every payment genuine" score?
2. Does the model beat it, and by how many rows?
3. Write one sentence for the product manager who saw "99.4 percent".

In the interview: "Why start with a baseline, and what does beating it prove?"

A3. What does this logistic output say? (warm-up, 5 minutes)

Seat: senior ML engineer. Kalpa's repeat model, one input only (invented):

```
intercept               -1.87
previous_purchases       0.69     (exp = 1.99)
```

1. Compute the chance of buying again for a member with 0 previous purchases, and for one with 4.
2. State the direction and the strength in one sentence each.
3. The member with 4 purchases scores about 0.71. Say what 0.71 means to Marketing.

In the interview: "What does a logistic output of 0.71 mean?"

A4. Who bought again? (medium, 10 minutes)

Seat: solution architect. Marketing asks for "members who buy again within 90 days" of a nudge sent on 31 March 2026. The window runs to 29 June. The order log (invented):

Member	Order date	Status
K1	12 Feb	delivered
K1	20 Apr	returned
K2	3 Mar	delivered
K2	5 Apr	delivered
K3	15 Jan	delivered
K3	2 Jul	delivered
K4	25 Mar	delivered
K4	10 May	cancelled
K5	1 Apr	delivered

A teammate's query:

```
        select customer_id,
               count(*) as orders_total,
               case when count(*) > 1 then 1 else 0 end as repeat_90d
        from orders
        group by customer_id;
```

1. Before computing anything, what one question do you ask Marketing?
2. Name the two lines that leak and say what each borrows from the future.
3. With your answer to question 1 assumed (state it), compute each member's honest flag.
4. What breaks if this query runs on 15 May, before the window closes?

In the interview: "Where is the 90-day repeat flag computed, from which table, as of which date?"

A5. Minutes per item at a dark store (medium, 10 minutes)

Seat: senior ML engineer, explaining to the store manager. Kalpa's Bengaluru dark store, five orders (invented):

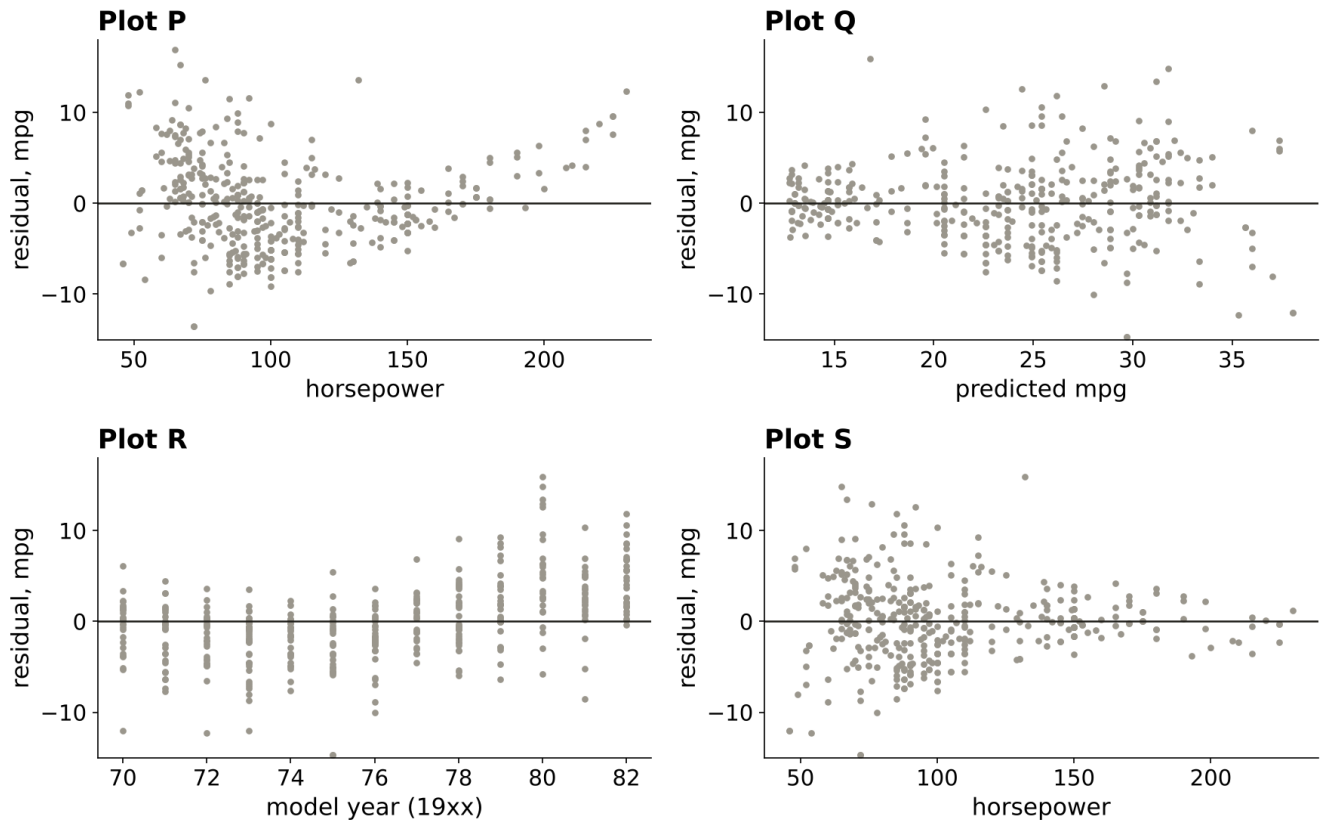
Items in the order	2	4	6	8	10
Packing minutes	5	7	10	11	14

1. Fit the least-squares line by hand: slope and intercept.
2. Read the slope in the manager's words.
3. Compute the sum of squared residuals for your line and for the flat mean line.
4. The manager asks for the packing time of a 30-item order. What do you say?

In the interview: "Explain one coefficient of a model you built to someone who has never seen a regression."

A6. Four residual plots, four confessions (medium, 10 minutes)

Seat: QA and tester. Four residual plots from UCI Auto MPG's 392 cars: plot P is the straight line on horsepower; Q, R and S are the line with horsepower squared, plotted against different things.



Match each plot to one diagnosis; one diagnosis is spare.

	Diagnosis
a	an input the model does not use still carries signal
b	the misses grow with the prediction
c	the straight line missed a curve
d	the test set was peeked at
e	centred on zero across the range: no shape left for this input

1. Post your match as P?, Q?, R?, S?.
2. For the plot that says "an input is missing", name the input and what the fix would be.

In the interview: "You fit a regression and the residual plot is not flat. What do you look for, in order?"

A7. The 0.906 in the status update (medium, 10 minutes)

Seat: senior ML engineer. A junior at the bank's analytics team (UCI Bank Marketing) posts:

```
validation accuracy (seed 42)          0.8971
tried 25 column lists on the test set
best test accuracy                      0.9060   <- reporting this one
```

1. What is 0.9060 now an estimate of?
2. What does the team report today, and what does it do before the next test score?
3. Calling nobody scores 0.8874. By how much did the reported edge grow over the validation edge?

In the interview: "Train, validation, test: what is each for, and which is spent once?"

A8. Booking.com: a regression or a classification? (interview grade, 15 minutes)

Seat: product manager with the senior ML engineer. Booking.com's KDD 2019 paper describes "a regression predicting the length of a stay, turned in to a multiclass classification problem", and prefers set-ups "where simple models can do significantly better than trivial models". Suppose the two set-ups scored as below on validation (the numbers are invented for this problem).

Set-up	Trivial baseline	Simple model
regression: nights of stay	mean: off by 1.9 nights on average	linear: off by 1.7 nights
classification: 1, 2, 3, 4 or more nights	majority class: right 41 percent	logistic: right 58 percent

1. What do you ask the product team before you compare the two rows?
2. By Booking's own rule, which set-up looks stronger, and how do you compare gains that are in different units?
3. Name one fact that would make you keep the regression anyway.

In the interview: "Marketing wants a model; walk me through how you decide what exactly it predicts."

A9. The model ties the baseline on Thursday (interview grade, 15 minutes)

Seat: senior ML engineer, talking to the business owner. The bank's validation slice, from class:

Option	Accuracy on 8,238 calls
call nobody	0.8874
one-line rule: call clients whose last campaign succeeded	0.8988
logistic model on 13 columns	0.8971

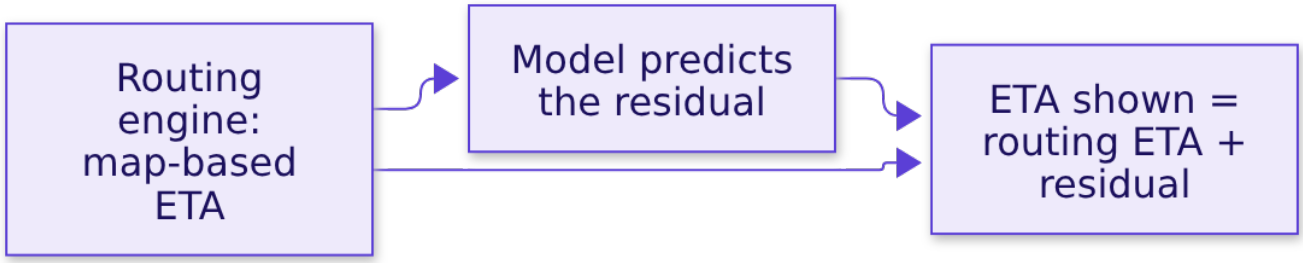
Marketing wants the score by Friday.

1. What do you ask before you answer?
2. What do you ship on Friday, and what do you say about the model?
3. What two numbers sit side by side in your message?
4. What result would make you ship the model instead?

In the interview: "Marketing wants the score by Friday and your model only ties the baseline. What do you tell them?"

A10. Uber's ETA, Kalpa's delivery promise (interview grade, 15 minutes)

Seat: solution architect. Uber's DeepETA post (10 February 2022): "We then use machine learning to predict the residual between the routing engine ETA and real-world observed outcomes." The new model had to improve mean absolute error "significantly over the incumbent XGBoost model" and "return an ETA within a few milliseconds at most".



1. For the residual model, name the input, the target and the prediction time.
2. What is the baseline the model must beat, and on which two measures?
3. Kalpa Logistics wants a delivery promise at checkout. Frame it the same way: the rule that stays, the target, and the baseline.
4. Name one change upstream that would silently change the residual model's target.

In the interview: "How would you predict delivery time for a quick-commerce order at checkout?"

Part B: eighteen self-check questions

Post one line in this shape (the letters here are an example, not the key): `1x 2x 3x ... 18x`, with several letters where a question asks for all that apply.

B1. Which question needs a trained model?

- a) What share of members bought twice last quarter?
- b) Flag every order above Rs 50,000 from a week-old member
- c) Which members will lapse in the next sixty days?
- d) What was the median delivery time in Pune last month?

B2. The validation slice is for:

- a) training the final model on more rows than train holds
- b) comparing models and settings, as often as needed
- c) reporting the final score once the model is frozen
- d) checking that the seed gives the same split each time

B3. A churn flag is 92 to 8 on validation (invented). "Nobody churns" scores:

- a) 8 percent
- b) 50 percent
- c) 100 percent
- d) 92 percent

B4. A logistic model scores a member 0.71. The right reading:

- a) of 100 similar members, about 71 buy again, if calibrated
- b) this member will buy again; the model is 71 percent sure
- c) the next basket will be 71 percent of the last basket
- d) 71 percent of all members will buy again within the window

B5. A linear regression gives visits a coefficient of Rs 120. For Marketing:

- a) each visit causes Rs 120 more spend, so push for one more visit
- b) one more visit goes with about Rs 120 more spend, in this data
- c) visits explain Rs 120 of the average member's monthly spend
- d) a member with zero visits still spends Rs 120 every month

B6. The least-squares line is "best" because it:

- a) passes through as many of the points as possible
- b) joins the first and the last point of the data
- c) leaves the smallest sum of squared residuals
- d) has the smallest slope that still fits the data

B7. Residuals against horsepower are positive at both ends and negative in the middle. This says:

- a) a straight line misses a curve in the data
- b) the data has outliers to remove before fitting

- c) the model is overfitting its training rows
- d) horsepower has no link with mileage at all

B8. After choosing a setting by its test score, the test score is now:

- a) still unbiased, since no test row was trained on
- b) pessimistic, since the model is less finely tuned
- c) equal to the validation score in expectation
- d) optimistic, because the choice was made on it

B9. A nudge goes out on 30 June. Which inputs may the model read? Select all that apply.

- a) orders placed on or before 30 June
- b) whether the member ordered in July
- c) days since the last order before 30 June
- d) the member's spend in the 90 days after 30 June

B10. Which make a split reproducible and honest? Select all that apply.

- a) a stated random seed
- b) stratifying on the target
- c) scoring the test slice after every change
- d) cutting the test slice before any modelling

B11. Which belong beside a classifier's validation accuracy? Select all that apply.

- a) the majority class's accuracy
- b) a one-line business rule's accuracy
- c) the model's accuracy on its own training rows
- d) the best test score from earlier runs

B12. "The bank's model, at 89.7 percent accuracy, has learned a lot about who says yes."

- a) True, because 89.7 percent is near ninety
- b) True, because it is right on most calls
- c) False, since calling nobody scores 88.7
- d) False, because above 85 means overfitting

B13. "A logistic coefficient is the change in probability per unit of input."

- a) True, because it is fitted like a line's slope
- b) True, because the model outputs a probability
- c) False, because logistic models have no coefficients
- d) False, it moves the log-odds per unit

B14. M4 ordered on 25 April and again on 15 October. "With a 30 June cutoff, M4's 90-day repeat flag is 1."

- a) True, because M4 did buy again
- b) True, because any later order counts
- c) False, the order falls after the window
- d) False, because M4 had one earlier order only

B15. Assertion: the test slice should be scored only once. Reason: every choice made with a score makes that score optimistic.

- a) A and R true; R explains A
- b) A and R true; R does not explain A
- c) A true; R false
- d) A false; R true

B16. Assertion: for spend = 180 + 120 x visits, Rs 180 is a safe forecast for a member with zero visits. Reason: the least-squares line passes through the mean of x and the mean of y.

- a) A and R true; R explains A
- b) A and R true; R does not explain A
- c) A true; R false
- d) A false; R true

B17. Match each slice to its job (one job is spare): 1 train, 2 validation, 3 test. Jobs: p) learn the coefficients; q) choose between models; r) report once at the end; s) tune the seed until the score is best.

- a) 1q 2p 3r
- b) 1p 2q 3r
- c) 1p 2r 3q
- d) 1r 2q 3p

B18. Match each residual pattern to what it says (one meaning is spare): 1 a bow; 2 a fan; 3 a trend against a column the model leaves out. Meanings: p) a missing curve; q) the misses grow with the prediction; r) an input left out; s) nothing left to learn.

- a) 1p 2q 3r
- b) 1q 2p 3r
- c) 1p 2s 3r
- d) 1r 2q 3p

STOP. Part C starts here: the key and the model solutions

The key

Item	Key	Item	Key	Item	Key
B1	c	B7	a	B13	d
B2	b	B8	d	B14	c
B3	d	B9	a, c	B15	a
B4	a	B10	a, b, d	B16	d
B5	b	B11	a, b	B17	b
B6	c	B12	c	B18	a

Part A answers, in one line each: A1 query, prediction (classification), rule, prediction (regression), query. A2 0.995; no, 10 rows short. A3 0.13 and 0.71. A4 ask whether a returned order counts; K1 0 or 1 by that answer, K2 1, K3 0, K4 0, K5 outside the table. A5 slope 1.1 minutes per item, intercept 2.8; 0.8 against 49.2. A6 Pc, Qb, Ra, Se. A7 a second validation score; the edge grew from 0.97 to 1.86 points. A8 the decision first; classification by its gain over trivial. A9 the clarifying question, then ship the rule. A10 the routing engine is the rule that stays.

Model solutions to Part A

A1. Questions 1 and 5 are queries: the answer already sits in the data. Question 3 is a rule: someone wrote the decision as a condition. Question 2 is a prediction, classification (will they miss a payment); question 4 is a prediction, regression (how much). Question 3 turns into a prediction problem when the business wants to approve some no-PAN applicants who will repay, so it needs a score for who is safe, for example once a regulator allows another identity document. *What the interviewer listens for:* the three tests named in order, and regression against classification read from "how much" against "will they". *The common wrong answer:* "all five, since we have the data"; the data makes a dataset, the decision makes the problem.

A2. "Call every payment genuine" scores 9,950 / 10,000 = 0.995. The model's 0.994 is right on 9,940 rows and wrong on 60; the baseline is wrong on 50, so the model is 10 rows worse than doing nothing. For the product manager: "Our model is right 99.4 percent of the time, and saying 'genuine' to everything is right 99.5 percent of the time, so on this measure it has not yet learned anything useful." *Listens for:* the baseline computed before any judgement. *Common wrong answer:* "99.4 is excellent"; an absolute score has no meaning without its bar.

A3. At 0 purchases, `z = -1.87`, `p = 1 / (1 + exp(1.87)) = 0.13`. At 4, `z = -1.87 + 4 x 0.69 = 0.89`, `p = 0.71`. Direction: positive, more previous purchases go with a higher chance of buying again. Strength: each extra purchase multiplies the odds by about 2. For Marketing: "Of 100 members who look like this one, about 71 buy again within 90 days, if the model is calibrated." *Listens for:* odds, never "adds 0.69 to the probability". *Common wrong answer:* "this member is 71 percent certain to buy", which turns a group rate into a promise.

A4. Ask first: "Does a returned or a cancelled order count as buying again?" The leaking lines: `count(*) as orders_total` counts orders after 31 March, so the feature includes the target's own purchase; `case when count(*) > 1` defines the target on the whole log with no cutoff and no window.

Assume delivered orders only count: K1 0 (its April order was returned), K2 1 (5 April is inside the window), K3 0 (2 July is after 29 June), K4 0 (its May order was cancelled). K5 has no order on or before 31 March, so it cannot "buy again" and does not belong in the table at all; a member's first order is a different question. Counting returns as purchases makes K1 a 1. Run on 15 May, the window is still open, so members who will buy in June get a false 0. *Listens for:* the clarifying question, a frozen reference date, and both sides built from it. *Common wrong answer:* "compute it fresh from the latest table each run"; fresh is right for scoring inputs, wrong for the target.

A5. Mean items 6, mean minutes 9.4. `dx`: -4, -2, 0, 2, 4; `dy`: -4.4, -2.4, 0.6, 1.6, 4.6. `slope = 44 / 40 = 1.1` minutes per item; `intercept = 9.4 - 1.1 x 6 = 2.8` minutes. For the manager: "Each extra item adds about one minute of packing, on top of about three minutes for any order." Predictions 5.0, 7.2, 9.4, 11.6, 13.8; residuals 0, -0.2, 0.6, -0.6, 0.2; sum of squares 0.8 against 49.2 for the mean line. A 30-item order is three times the largest order seen, so the honest answer is "about 36 minutes if the line holds, and we have never seen an order that big; let me check orders over 10 items first." *Listens for:* units, and refusing to extrapolate silently. *Common wrong answer:* quoting 35.8 minutes as a fact.

A6. P is c, the bow of a straight line on a curve. Q is b, the misses fan out as predicted mileage rises. R is a: the residuals climb with model year, from about -2 mpg in the mid-1970s to about +5 in 1980 to 1982, so newer cars beat the model; add `model_year` as an input. S is e: centred on zero at every horsepower once the squared term is in (its spread narrows at high horsepower, where there are few cars). Spare: d, a peek never shows in a residual plot. *Listens for:* the pattern named before the fix. *Common wrong answer:* reading Q's fan as "the model is wrong everywhere".

A7. 0.9060 is now a second validation score: it was used to choose among 25 column lists, so it is optimistic by however much the choosing bought. Report the validation score (0.8971, seed 42) today, and before the next test score cut a fresh held-out slice that no choice has touched. The validation edge is 0.8971 - 0.8874 = 0.97 points; the reported edge is 0.9060 - 0.8874 = 1.86 points, nearly double. *Listens for:* "the test set is spent once" with its reason. *Common wrong answer:* "it is still unbiased because no test row was trained on"; choosing by it leaks the same way.

A8. Ask first: "What decision does the prediction feed: a price, a room allocation, an email?" A decision that needs the exact night count favours regression; one that only needs a bucket favours classification. On Booking's rule, compare each simple model's gain over its own trivial baseline: regression cuts the error by 0.2 of 1.9 nights, about 11 percent; classification lifts accuracy from 41 to 58 percent, 17 points, which also removes 29 percent of the baseline's errors (59 to 42). The classification set-up shows the bigger gain over trivial, so it is preferred. Keep the regression if the decision needs the night count itself, for example pricing a long-stay discount by the night. *Listens for:* the decision before the metric, and gains read against each set-up's own baseline. *Common wrong answer:* comparing 1.7 nights with 58 percent directly.

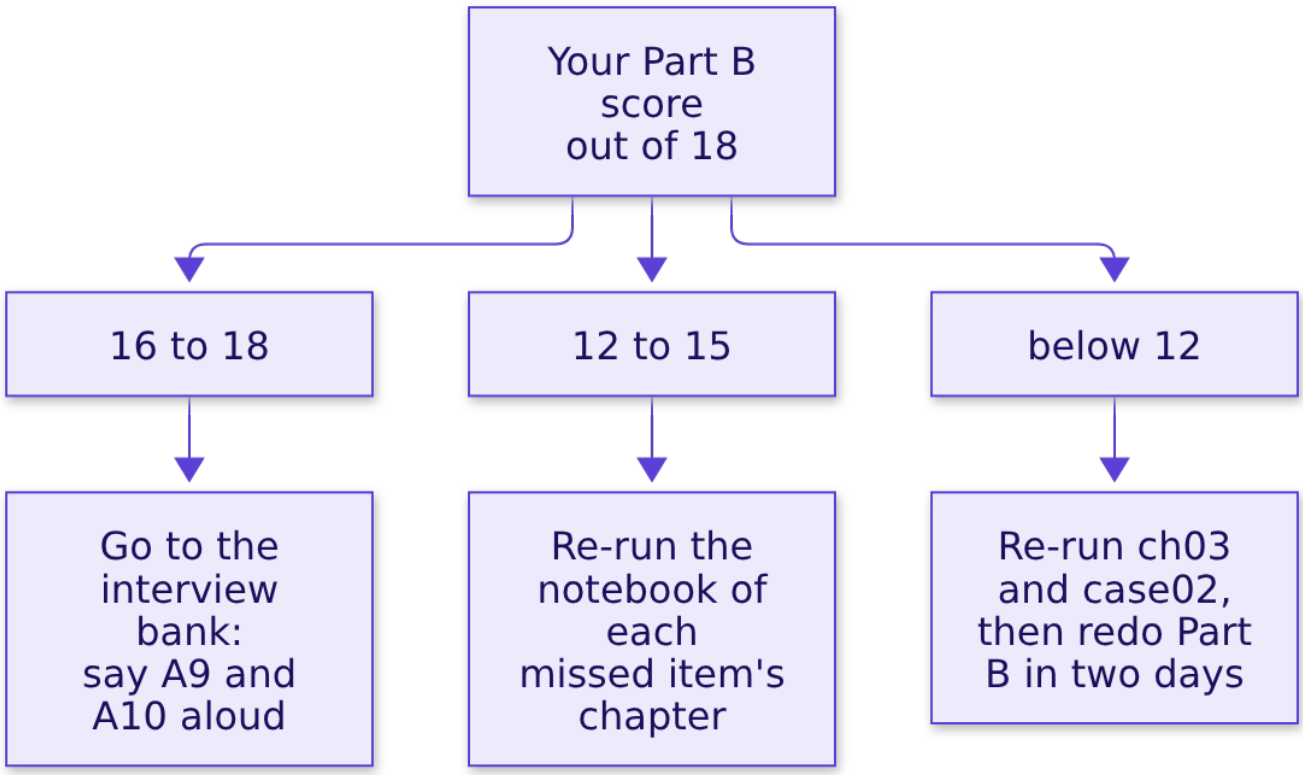
A9. Ask first: "Ties on what measure, and on which slice?" Today's measure is accuracy on the validation slice; Tuesday shows accuracy flatters on 89 to 11 data. Ship the one-line rule on Friday: it scores 0.8988 against the model's 0.8971, costs nothing to run and is easy to explain. Say: "The model has not yet beaten a rule the campaign team can run today; I am asking for the week to test it on the measure that matches the campaign's cost." Side by side: 0.8874 for calling nobody and 0.8988 for the rule, with the model's 0.8971 beside them. Ship the model when it beats the rule on validation by more than the noise of the slice, on the measure that matches the cost of a wasted call against a missed yes. *Listens for:* the clarifying question, the baseline and the rule named, a date for the next check. *Common wrong answer:* "ship it, it is no worse"; a model that ties costs maintenance and trust for nothing.

A10. Input: the routing engine's ETA and the trip's features known when the request is made; target: the actual arrival time minus the routing engine's ETA; prediction time: the moment the ETA is shown. The baseline is the incumbent XGBoost model, and the new one had to beat it on mean absolute error and answer within a few milliseconds. For Kalpa Logistics: the rule that stays is the distance-and-slot promise the checkout already shows; the target is the actual delivery time minus that promise; the baseline is the promise itself (a residual of zero) and then a simple regression on the residual. Upstream risk: if the routing engine or the promise rule changes, every residual changes meaning overnight, so the model must be retrained with the rule's version recorded. *Listens for:* the residual framing and the latency constraint. *Common wrong answer:* predicting delivery minutes from scratch and throwing away a rule that already works.

One line on every option of Part B

Item	a	b	c	d
B1	a query: the answer is in the data	a rule: the condition is written	key: future, unknown, decision moves	a query over last month
B2	that is train's job	key	that is the test slice's job	the seed is fixed in code, no slice needed
B3	the share of the minority	a coin flip, which ignores the 92	never wrong is impossible here	key: 92 of 100 are "no churn"
B4	key: a group rate, if calibrated	turns a rate into a promise	confuses probability with basket size	confuses one member's score with the base rate
B5	reads association as cause	key	a coefficient is a slope, not a share	the intercept is beyond the data
B6	lines rarely pass through points	the eye's line, beaten by least squares	key	the slope is whatever fits; size is no criterion
B7	key	outliers scatter, they do not bow	overfitting shows in train against validation	a bow says there is a link, curved
B8	no training on rows is needed to leak	choosing makes it higher, not lower	it is now biased upward	key
B9	key	July is after the cut	key	the window is the target
B10	key	key	that spends the test many times	key
B11	key	key	training accuracy flatters the model	earlier test scores were peeks
B12	near ninety means nothing alone	most calls are "no" anyway	key	high accuracy is not overfitting by itself
B13	it is fitted on log-odds	the output is a probability; the coefficient is not	the model has one per input	key
B14	it bought, outside the window	the window is 90 days, not forever	key	earlier orders do not define the target
B15	key	R is exactly why A holds	R is true	A is true
B16	A is false	A is false	A is false, R true	key
B17	train and validation swapped	key	validation and test swapped	train and test swapped
B18	key	bow and fan swapped	a fan is not "nothing left"	bow and trend swapped

What should you revisit?

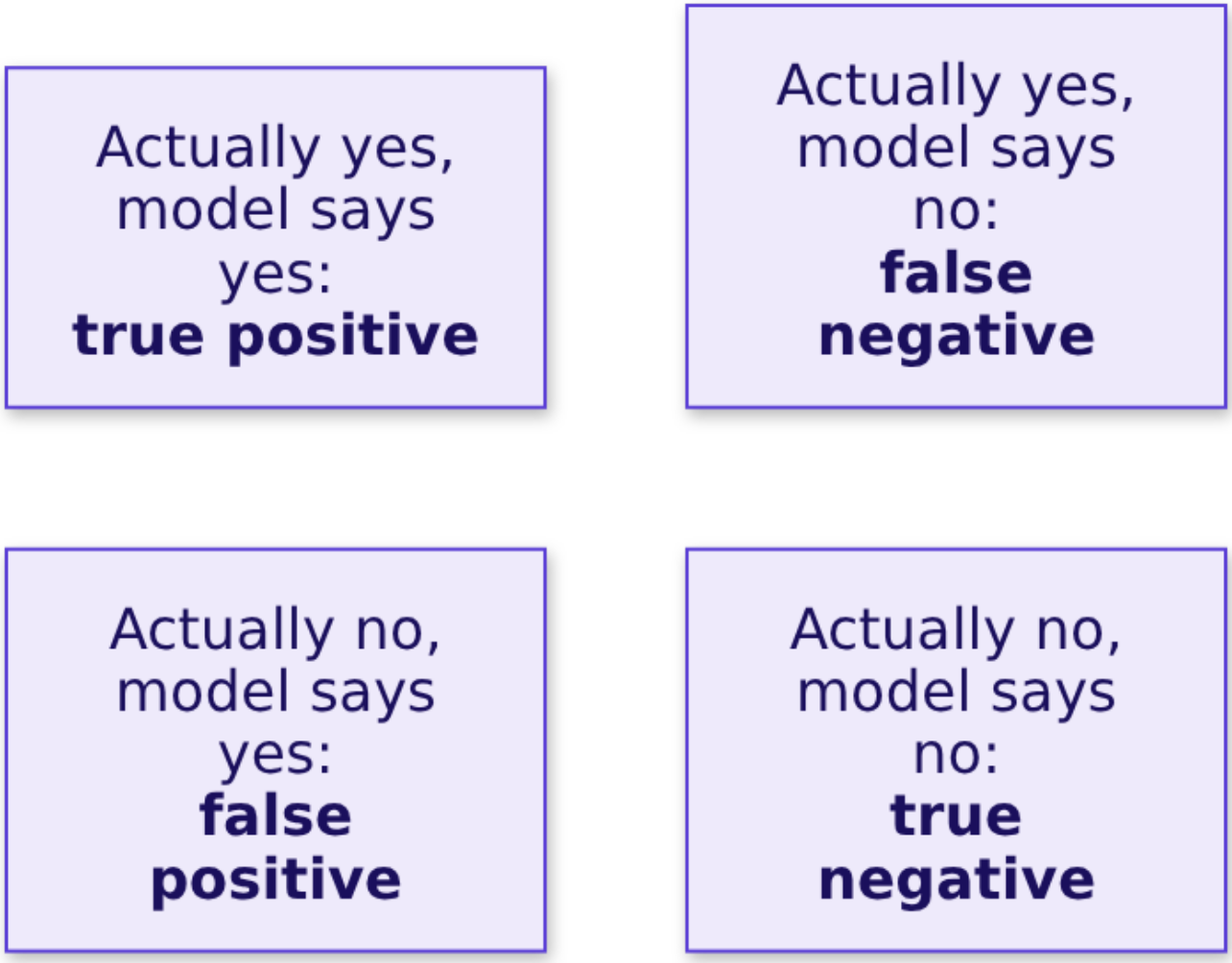


Missed	Revisit
B1, B9, B14	chapters 1 and 2, notebooks ch01 and ch02
B2, B8, B10, B15, B17	chapter 3 and case 3, notebooks ch03 and case03
B3, B11, B12	case 2, notebook case02
B5, B6, B16	chapter 4, notebook ch04, and the explorable
B4, B13	chapter 5, notebook ch05
B7, B18	chapter 6, notebook ch06

Where do you read further?

- UCI Bank Marketing, the dataset page and its variable notes: https://archive.ics.uci.edu/dataset/222/bank+marketing (verified 28 October 2026)
- UCI Auto MPG, the dataset page: https://archive.ics.uci.edu/dataset/9/auto+mpg (verified 28 October 2026)
- Google, Rules of Machine Learning, rules 1, 4 and 14: https://developers.google.com/machine-learning/guides/rules-of-ml (verified 28 October 2026)
- scikit-learn 1.9.1 user guide, linear models: ordinary least squares and logistic regression: https://scikit-learn.org/stable/modules/linear_model.html (verified 28 October 2026)
- StatQuest video index: "The Essence of Linear Regression" and "Logistic Regression": https://statquest.org/video_index.html (verified 28 October 2026)

Tomorrow's ten-minute pre-read: what accuracy hides



Term	One line
Confusion matrix	the four boxes above, counted on the validation slice
Precision	of the clients the model said yes to, the share who said yes
Recall	of the clients who said yes, the share the model found
Threshold	the probability above which the model's answer becomes "call"
Base rate	the share of yes before any model: 0.113 at the bank